Individual Coursework Submission Form

Specialist Masters Programme

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Q2.

a.) Figure 2.1 shows the level of each variable, from which it can be seen that lgoldq, lnaturalgasq, lcrudeoilq, and lcopperq have clear deterministic trend and are obviously non-stationary.

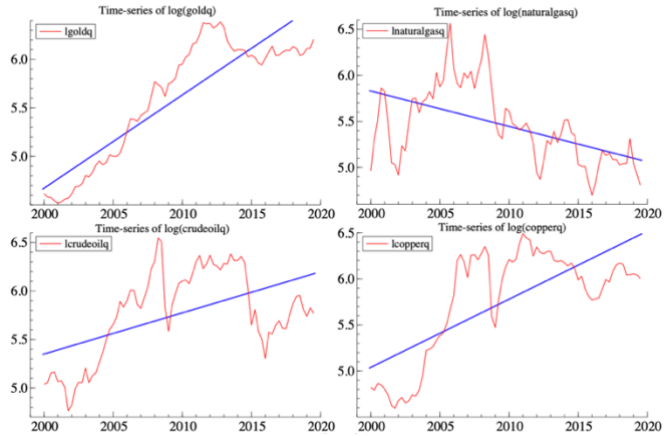

Figure 2.1

To further investigate whether they are trend-stationary or difference-stationary, deterministic trend needs to be removed. Table 2.1 shows the result of simple linear regression via OLS of each variable.

$lgoldq = \alpha + \beta * t + \epsilon_t$	**coefficient**	**t-value**	**t-prob**
Constant	4.64765	74.2	0.0000
Trend	0.0240396	17.7	0.0000
$lnaturalgasq = \alpha + \beta * t + \epsilon_t$	**coefficient**	**t-value**	**t-prob**
Constant	5.83796	70.4	0.0000
Trend	-0.00958401	-5.32	0.0000
$lcrudeoilq = \alpha + \beta * t + \epsilon_t$	**coefficient**	**t-value**	**t-prob**
Constant	5.02048	54.3	0.0000
Trend	0.0185032	9.21	0.0000

Table 2.1

From Table 2.1, it can be seen that all series have statistically significant deterministic trend. If we plot residuals ε_t from regressions above, we can obtain de-trended series for each variable as showed in Figure 2.2 below.

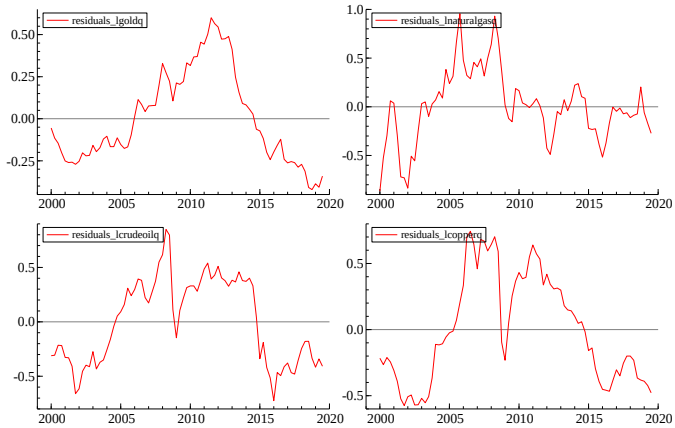

Figure 2.2

It is clear that after removing the deterministic trend for each series, series are still dominated by a stochastic trend, which means all series are not trend stationary. To make the series stationary, 1st-order difference should be applied.

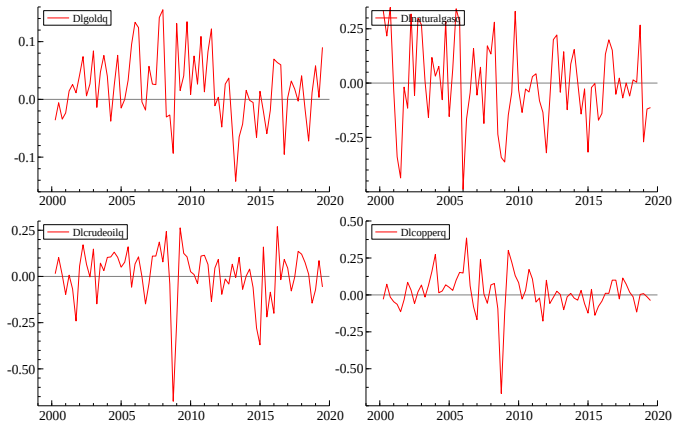

Figure 2.3

Figure 2.3 illustrates that the 1st-order difference of each series seems more stationary. Thus, lgoldq, lnaturalgasq, lcrudeoilq and lcopperq seem to be difference stationary. However, unit root tests are required to prove this, as Holden and Perman (1994) suggests, there is an association between a series possessing a "unit root" and being "difference stationary". Unit root test will be constructed in *b.)* to statistically prove that lgoldq, lnaturalgasq, lcrudeoilq and lcopperq are difference stationary.

b.) The Dickey-Fuller test is used to test for unit roots. For ease of computation, the following regression in ***E.Q. 1*** is employed:

$$\Delta Y_t = \alpha + \beta * t + \theta * Y_{t-1} + u_t \quad \ldots\ldots\ldots\ldots\ldots \textbf{\textit{(E.Q. 1)}}$$

$H_0: \theta = 0 \implies$ the time series contains a unit root, i.e. $Y_t \sim I(1)$;

$H_1: \theta \neq 0 \implies$ the time series is stationary, i.e. $Y_t \sim I(0)$.

	Dickey-Fuller Test on Δlgoldq				
Independent Variable	Coefficient	Std.Error	t-value	t-prob	Part.R^2
Constant	0.0944746	0.1148	0.823	0.4132	0.0089
Trend	-8.83349e-05	0.0006717	-0.132	0.8957	0.0002
$lgoldq_{t-1}$	-0.0125892	0.02470	-0.510	0.6118	0.0035
AR 1-5 test:	F (5, 70) = 1.1780 [0.3288]				
ARCH 1-4 test:	F(4,70) = 1.1627 [0.3348]				
Normality test:	$Chi^2(2)$ = 0.66428[0.7174]				
Hetero test:	F(4,73) = 1.9122 [0.1175]				
Hetero-X test:	F(5,72) = 1.7831 [0.1272]				
RESET23 test:	F(2,73) = 4.9669 [0.0095]**				

Table 2.2

	Dickey-Fuller Test on ΔΔlgoldq				
Independent Variable	Coefficient	Std.Error	t-value	t-prob	Part.R^2
Constant	0.0159079	0.006896	2.31	0.0238	0.0663
$\Delta lgoldq_{t-1}$	-0.732834	0.1117	-6.56	0.0000	0.3648
AR 1-5 test:	F(5,70) = 0.19633 [0.9629]				
ARCH 1-4 test:	F(4,69) = 0.79458 [0.5327]				
Normality test:	$Chi^2(2)$ = 0.15022[0.9276]				
Hetero test:	F(2,74) = 1.0439 [0.3572]				
Hetero-X test:	F(2,74) = 1.0439 [0.3572]				
RESET23 test:	F(2,73) = 0.53222 [0.5896]				

Table 2.3

After applying DF-test to $\Delta lgoadq$, Table 2.2 suggests that the model is almost well-specified, as it has only problem with functional form and no issue regarding to auto-correlation of residuals, which means augmented DF-test is not necessary. From Table 2.2, t-value of β and θ in first differencing model are **-0.132** and **-0.510** respectively. By Holden and Perman (1994), critical values from the standard normal are appropriate for β when it is non-zero, which means the trend in $\Delta lgoadq$ is statistically insignificant at 95% confidence level as t-value is inside the accept region of [-1.96, 1.96]. However, t-value of θ needs to be compared with the critical value from Dickey-Fuller table which is **-3.47** at 5% significance level in our sample. In this case, **-0.510** is larger than **-3.47**, thus, H_0 is not rejected in the first round. According to Holden and Perman (1994), to evaluate the order of integration of a series, further

differencing is required. Therefore, we evolve to the model in ***E.Q. 2***:

$$\Delta\Delta Y_t = \alpha + \theta * \Delta Y_{t-1} + u_t \quad \ldots\ldots\ldots\ldots \textbf{\textit{(E.Q. 2)}}$$

Where H_0 and H_1 remains the same as in ***E.Q. 1.***

Applying DF-test again to $\Delta\Delta lgoldq$, from Table 2.3, it is clear that in this model, t-value of θ is **-6.56** which is less than the critical value from DF-table at **-3.47**, thus, H_0 is rejected. Therefore, $\Delta lgoldq_t \sim I(0)$ and $lgoldq_t \sim I(1)$. We can therefore assert that lgoldq is a difference stationary series.

	Dickey-Fuller Test on Δlnaturalgasq				
Independent Variable	Coefficient	Std.Error	t-value	t-prob	Part.R^2
Constant	1.05707	0.3378	3.13	0.0025	0.1155
Trend	-0.00275212	0.001064	-2.59	0.0117	0.0818
$lnaturalgasq_{t-1}$	-0.173456	0.05737	-3.02	0.0034	0.1086

AR 1-5 test:	F(5,70) = 3.2264 [0.0112]*
ARCH 1-4 test:	F(4,70) = 1.7064 [0.1583]
Normality test:	$Chi^2(2)$ = 0.89445[0.6394]
Hetero test:	F(4,73) = 2.8031 [0.0318]*
Hetero-X test:	F(5,72) = 2.2328 [0.0601]
RESET23 test:	F(2,73) = 1.2023 [0.3064]

Table 2.4

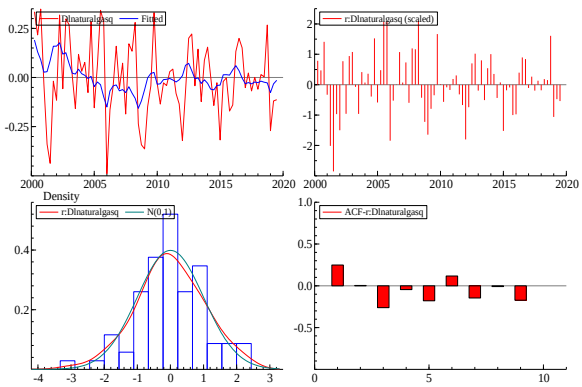

Figure 2.4

Table 2.4 shows the DF-test result for $\Delta lnaturalgasq$. As Table 2.4 and Figure 2.4 shows, there is a model misspecification regards autocorrelation of residuals and heterodasticity. As autocorrelation of residuals exists, lagged $\Delta lnaturalgasq$ needs to be introduced into the model, and therefore, ADF-test is necessary to get a less turbulence error term. The ADF-test result is in Table 2.5.

	Augmented Dickey-Fuller Test on Δlnaturalgasq				
Independent Variable	Coefficient	Std.Error	t-value	t-prob	Part.R^2
$Dlnaturalgasq_{t-1}$	0.229613	0.1251	1.84	0.0708	0.0479
$Dlnaturalgasq_{t-2}$	0.00883301	0.1155	0.0765	0.9393	0.0001
$Dlnaturalgasq_{t-3}$	-0.276494	0.1163	-2.38	0.0203	0.0778
$Dlnaturalgasq_{t-4}$	0.0615921	0.1193	0.516	0.6074	0.0040
Constant	0.816530	0.4053	2.01	0.0480	0.0571
Trend	-0.00184434	0.001195	-1.54	0.1275	0.0343
$lnaturalgasq_{t-1}$	-0.137020	0.06806	-2.01	0.0481	0.0570

AR 1-5 test:	F(5,62) = 4.3800 [0.0018]**
ARCH 1-4 test:	F(4,66) = 0.81548 [0.5198]
Normality test:	$Chi^2(2)$ = 0.25564[0.8800]
Hetero test:	F(12,61) = 1.4999 [0.1491]
Hetero-X test:	F(27,46) = 0.96516 [0.5290]
RESET23 test:	F(2,65) = 0.41401 [0.6627]

Table 2.5

	Augmented Dickey-Fuller Test on ΔΔlnaturalgasq				
Independent Variable	Coefficient	Std.Error	t-value	t-prob	Part.R^2
$DDlnaturalgasq_{t-1}$	0.618279	0.2171	2.85	0.0058	0.1080
$DDlnaturalgasq_{t-2}$	0.500995	0.1704	2.94	0.0045	0.1143
$DDlnaturalgasq_{t-3}$	0.163609	0.1452	1.13	0.2638	0.0186
$DDlnaturalgasq_{t-4}$	0.287241	0.1133	2.54	0.0136	0.0876
Constant	-0.00874533	0.02000	-0.437	0.6633	0.0028
$Dlnaturalgasq_{t-1}$	-1.45290	0.2568	-5.66	0.0000	0.3233

AR 1-5 test:	F(5,62) = 1.7615 [0.1340]
ARCH 1-4 test:	F(4,65) = 0.66007 [0.6220]
Normality test:	$Chi^2(2)$ = 0.15448[0.9257]
Hetero test:	F(10,62) = 0.49333 [0.8881]
Hetero-X test:	F(20,52) = 1.2582 [0.2489]
RESET23 test:	F(2,65) = 0.99565 [0.3750]

Table 2.6

From Table 2.6, it can be inferred that H_0 in ***E.Q. 2*** is rejected as t-value of θ in this regression is **-5.66** and less then the critical value from DF-table, which implies that $\Delta lnaturalgasq_t \sim I(0)$, and thus, $lnaturalgasq_t \sim I(1)$ and lnaturalgasq is difference stationary.

Table 2.7 shows the DF-test result on $\Delta lncrudeoilq$, and the model is not well-specified since there are issue regarding normality and heterodasticity. This is because there is a spike in the level which has not been modeled in as Figure 2.5 shows, which can be solved by further introducing a dummy variable to caputure this outlier.

Independent Variable	Dickey-Fuller Test on $\Delta lcrudeoilq$ Coefficient	Std.Error	t-value	t-prob	Part.R^2
Constant	0.386030	0.2262	1.71	0.0920	0.0374
Trend	8.85600e-07	0.0008597	0.00103	0.9992	0.0000
$lcrudeoil_{t-1}$	-0.0653537	0.04206	-1.55	0.1244	0.0312
AR 1-5 test:	F(5,70) = 1.5458 [0.1869]				
ARCH 1-4 test:	F(4,70) = 1.0574 [0.3842]				
Normality test:	$Chi^2(2)$ = 20.933[0.0000] **				
Hetero test:	F(4,73) = 1.2491 [0.2979]				
Hetero-X test:	F(5,72) = 2.7104 [0.0267]*				
RESET23 test:	F(2,73) = 2.9711 [0.0575]				

Table 2.7

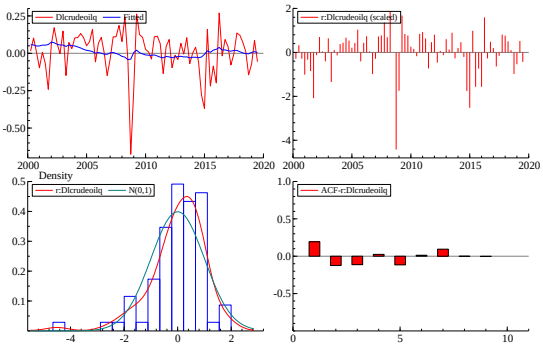

Figure 2.5

H_0 in ***E.Q. 2*** is rejected for $\Delta lncrudeoilq_t$ in the second round as Table 2.8 shows, t-value of θ (**-7.22**) is less then critical value of **-3.47** given by DF-table, which implies $\Delta lncrudeoilq_t \sim I(0)$, and therefore, $lcrudeoilq_t \sim I(1)$, lcrudeoilq is difference stationary.

Independent Variable	Augmented Dickey-Fuller Test on $\Delta\Delta lcrudeoilq$ Coefficient	Std.Error	t-value	t-prob	Part.R^2
Constant	0.386030	0.2262	1.71	0.0920	0.0374
$Dlcrudeoil_{t-1}$	-0.821312	0.1138	-7.22	0.0000	0.4100
AR 1-5 test:	F(5,70) = 1.2413 [0.2993]				
ARCH 1-4 test:	F(4,69) = 1.0110 [0.4079]				
Normality test:	$Chi^2(2)$ = 20.485[0.0000] **				
Hetero test:	F(2,74) = 1.5062 [0.2285]				
Hetero-X test:	F(2,74) = 1.5062 [0.2285]				
RESET23 test:	F(2,73) = 0.43060 [0.6518]				

Table 2.8

Graphic analysis of DF-test result for $\Delta lcopperq$ is in Figure 2.6. Similar to lcrudeoilq, there is also a misspecification in normality as an outlier exists in the level in Figure 2.6. Again, dummy variable can be set to capture this spike to obtain a well-specified model. Additionally, there is also an issue with autocorrelation in residuals, and thus, ADF-test is needed. Table 2.9 summaries the result of ADF-test.

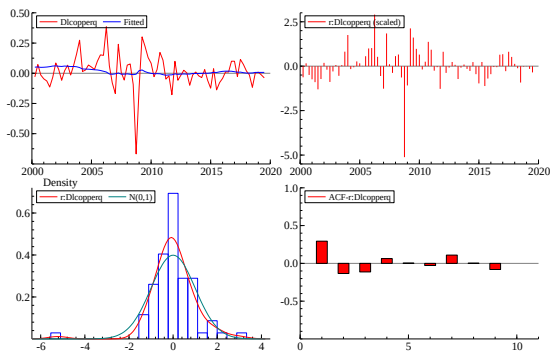

Figure 2.6

Augmented Dickey-Fuller Test on $\Delta lcopperq$	coefficient	t-value
$Dlcopperq_{t-1}$	0.379	3.12
$Dlcopperq_{t-2}$	-2.00747	-1.55
$Dlcopperq_{t-3}$	-0.0144777	-0.114
$Dlcopperq_{t-4}$	0.101375	0.835
Constant	0.326749	1.69
Trend	0.000324995	0.317
$lcopperq_{t-1}$	-0.0565899	**-1.47**
Augmented Dickey-Fuller Test on $\Delta\Delta lcopperq$	**coefficient**	**t-value**
$DDlcopperq_{t-1}$	0.252782	1.22
$DDlcopperq_{t-2}$	0.165761	0.0929
$DDlcopperq_{t-3}$	-0.0232431	-0.162
$DDlcopperq_{t-4}$	0.0825972	0.677
Constant	0.0152490	0.993
$Dlcopperq_{t-1}$	-0.880963	**-3.85**

Table 2.9

As Table 2.9 shows, θ is significanlt different from zero at the second round as its t-value of **-3.85** is less than the critical value of **-3.47**, at 95% confidence level. Thus, $\Delta lcopperq_t \sim I(0)$ *and* $lcopperq_t \sim I(1)$, and lcopperq is indeed difference stationary.

Q3.

a.) Table 3.1 shows the result of regression between lnaturalgasq and lcrudeoilq according to ADL (1, 1) model in ***E.Q. 3***. However, it can be seen from Figure 3.1 that ADL (1, 1) is not perfectly well-specified, as residuals are not iid and shows

serial correlation.

$$Y_t = \alpha + \beta * X_t + \gamma * X_{t-1} + \rho * Y_{t-1} + \varepsilon_t \quad \ldots\ldots\ldots\ldots \textbf{(E.Q. 3)}$$

	ADL(1, 1) of lnaturalgasq and lcrudeoilq				
Independent Variable	Coefficient	Std.Error	t-value	t-prob	Part.R^2
$lnaturalgasq_{t-1}$	0.906675	0.05008	18.1	0.0000	0.8158
Constant	0.606772	0.3480	1.74	0.0854	0.0395
lcrudeoilq	0.408556	0.1458	2.80	0.0065	0.0960
$lcrudeoilq_{t-1}$	-0.426380	0.1434	-2.97	0.0040	0.1067

AR 1-5 test:	F(5,69) = 3.5646 [0.0063]**
ARCH 1-4 test:	F(4,70) = 2.4246 [0.0561]
Normality test:	$Chi^2(2)$ = 0.092119[0.9550]
Hetero test:	F(6,71) = 1.9084 [0.0913]
Hetero-X test:	F(9,68) = 1.5072 [0.1632]
RESET23 test:	F(2,72) = 2.1808 [0.1204]

Table 3.1

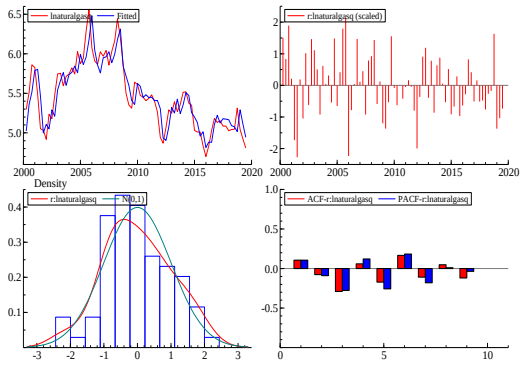

Figure 3.1

Imposing restrictions on ADL (1, 1), nine "nested" models is derived in Table 3.2.

Model	Restrictions	Test-statistic	Decision	AIC
Model 1	$\gamma = 0, \rho = 0$	F(2,74) = 166.65 [0.0000]**	Reject	
Model 2	$\rho\beta + \gamma = 0$	$Chi^2(1) = 1.4407[0.2300]$	Not reject	1.12960
Model 3	$\beta = 0$	F(1,74) = 7.8559 [0.0065]**	Reject	
Model 4	$\gamma = 0$	F(1,74) = 8.8348 [0.0040]**	Reject	
Model 5	$\rho = 0$	F(1,74) = 327.83 [0.0000]**	Reject	
Model 6	$\beta = 0, \gamma = 0$	F(2,74) = 4.4187 [0.0154]*	Reject	
Model 7	$\beta = 0, \rho = 0$	F(2,74) = 169.05 [0.0000]**	Reject	
Model 8	$\rho = 1, \beta = -\gamma$	$Chi^2(2) = 4.0002[0.1353]$	Not reject	-0.520815
Model 9	$\beta + \gamma + \rho = 1$	$Chi^2(1) = 3.2169[0.0729]$	Not reject	-0.507559

Table 3.2

From Table 3.2, it can be seen that Model 2, Model 8 and Model 9 are acceptable more parsimonious "nested" model of ADL (1, 1). According to AIC, the optimal "nested" model should be Model 8 as it has the minimum AIC, but autocorrelation in residuals still exists in Model 8. Thus, ADL model with higher lag might should be attempted, such as ADL (4, 4), to get a more well-specified model.

Q4.

a.) Assume there is a linear relationship between lnaturalgasq and lcrudeoilq, then a simple linear regression model as in ***E.Q. 4*** can be constructed:

$$lnaturalgasq_t = \alpha + lcrudeoil_t + \varepsilon_t \quad \ldots\ldots\ldots\ldots \textbf{(E.Q. 4)}$$

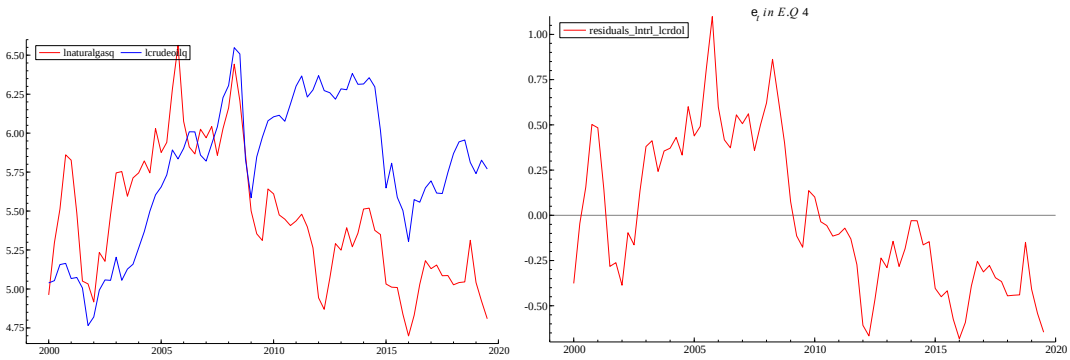

Figure 4.1

Figure 4.1 suggests that the residuals in *E.Q. 4* is clearly not a $I(0)$ process, which means the trend in lcrudeoilq cannot fully explain the trend in lnaturalgasq. Applying unit root test on ε_t from ***E.Q.4***, Table 4.1 can be obtained.

D-lag of ϵ_t	t-adf	AIC
4	-2.386	-3.496
3	-2.195	-3.505
2	-2.914	-3.466
1	-3.021	-3.492
0	-2.540	-3.469
D-lag of Diff(ϵ_t)	**t-adf**	**AIC**
4	-5.837**	-3.527
3	-4.867**	-3.441
2	-6.874**	-3.463
1	-6.140**	-3.375
0	-7.620**	-3.395

Table 4.1

According to Dickey-Fuller table, this suggests that the first difference of ε_t is stationary and ε_t itself is non-stationary, which implies $\varepsilon_t \sim I(1)$, and thus, lnaturalgasq and lcrudeoilq is spuriously correlated rather than cointegrated.

b.) According to Yule (1926), spurious regression happens when a statistically significant linear relationship is found between two actually independent time series. Spurious correlation exists when two variables are apparently correlated only because each is correlated with an omitted variable. As ***E.Q. 4*** indicates, a linear combination of two spuriously correlated variables is a non-stationary series, which means the residuals in the linear combination contains a non-I (0) process, and this may due to omitted variables (Ferson et al., 2003). Unlike spurious regression, cointegration regression refers to the scenario where the linear combination of two non-stationary series is a I (0) process. This is because when, for example, two I (1) processes can cancel out the disturbance in residuals when combine together. Levels of many economic variables themselves are non-stationary and their differences are stationary. However, standard inference procedures rely on the perquisition of stationary process. It can lead to spurious regression if we regress the non-stationary levels directly. Thus, differencing is required as differencing attenuates large positive residual serial correlation and reduces those series to a non-integrated form, where standard inference procedures can be applied. Differencing is a feasible alternative to avoid spurious regression from a statistical point of view, however, economic theory was mostly formulated in levels rather than in differenced series (Hendry, 2004).

c.) Since $\varepsilon_t \sim I(1)$ from ***E.Q. 4***, ADL (1, 1) in ***E.Q.5*** can be applied to lnaturalgasq to check whether residuals are stationary or not:

$$lnaturalgasq_t = \alpha + \beta lcrudeoil_t + \gamma lcrudeoilq_{t-1} + \rho lnaturalgasq_{t-1} + \varepsilon_t \quad \text{.....} \quad \textbf{(E.Q. 5)}$$

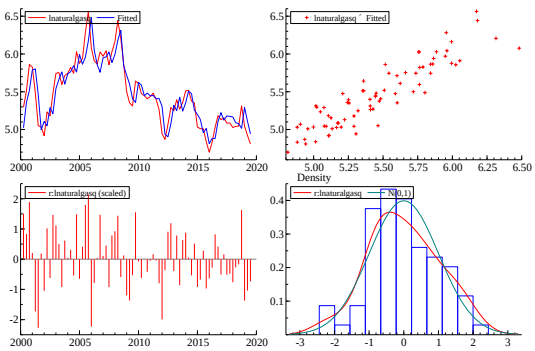

Figure 4.2

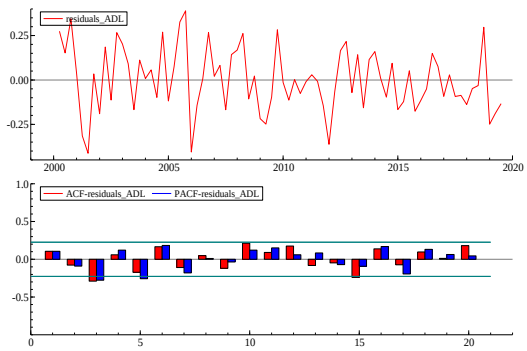

Figure 4.3

Figure 4.2 and Figure 4.3 shows that ADL (1, 1) on lnaturalgasq is not well-specified as autocorrelation exists. However, residuals in ADL (1, 1) are stationary and has order of integration 1 according to unit root test. An error correction model (Model 9) as in ***E.Q. 6*** nested in this ADL (1, 1) can be constructed according to Engle and Granger:

$$\Delta lnaturalgasq_t = \beta * lcrudeoilq_t + \alpha * \hat{u}_{t-1} + v_t, \text{ where } \hat{u}_{t-1} = Y_{t-1} - X_{t-1} \quad \ldots\ldots\ldots\ldots \quad \textbf{\textit{(E.Q. 6)}}$$

Apply dynamic analysis on ECM in ***E.Q. 6***, we can get:

$$ECM = lnaturalgasq - 6.50173 + 0.190994 * lcrudeoilq$$
$$WALD\ test: Chi^{2(1)} = 0.131967\ \ [0.7164]$$

The unit root t-test statistic for this ECM is **-1.8637**, which is larger than critical value from Dicky-Fuller table. Thus, according to Engle and Granger, the null hypothesis of no cointegration exists cannot be reject, which is in line with our finding in Figure 4.1. However, as we applied ADL (1, 1) and ECM to lnaturalgasq and lcrudeoilq, in which additional lags and independent variables are imposed at the risk of data mining. According to Hendry (2004), this may amplify the spurious regression problem, as data mining can intentionally lead to more persistent ones to be chosen.

Q5.

a.) To choose which lags in each variable should be included, ADF-test can be used and the lag with minimum AIC will be selected, as Table 5.1 shows.

D-lag of diff(lnaturalgasq)	AIC	Decision
4	-3.452	Choosed
3	-3.379	
2	-3.404	
1	-3.319	
0	-3.340	
D-lag of Diff(lcrudeoilq)	**AIC**	**Decision**
4	-3.757	
3	-3.743	
2	-3.770	
1	-3.789	Choosed
0	-3.779	
D-lag of diff(lcopperq)	**AIC**	Decision
4	-4.048	
3	-4.067	
2	-4.091	
1	-4.118	Choosed
0	-4.072	

Table 5.1

The model used to test Granger causality is in ***E.Q. 7*** . Difference used to ensure

series in this regression are stationary:

$$\Delta Y_t = \sum_{i=1}^{n} \beta_{1,i} * \Delta Y_{t-i} + \sum_{i=1}^{n} \beta_{2,i} * \Delta lgoldq_{t-i} + u_i \quad \ldots\ldots\ldots\ldots \textbf{\textit{(E.Q. 7)}}$$

$$H_0: \beta_{2,1} = \beta_{2,2} = \beta_{2,3} = \cdots = \beta_{2,i} = 0 \quad \Longrightarrow X_t \text{ fail to Granger-cause } Y_t;$$

$$H_1: \beta_{2,1} \neq \beta_{2,2} \neq \beta_{2,3} \neq \cdots \neq \beta_{2,i} \neq 0 \quad \Longrightarrow X_t \text{ Granger-cause } Y_t.$$

Table 5.2 summaries the result of Granger causality test. H_0 is rejected at 95% confidence level if p-value, i.e. t-prob of all coefficients of $\Delta lgoldq_{t-i}$ in the table are jointly less than **0.05**.

Granger Causality Test with $\Delta lnaturalgasq$	**t-value**	**t-prob**	**Decision**
$\text{Dlnaturalgasq}_{t-1}$	-1.49	0.1422	
$\text{Dlnaturalgasq}_{t-2}$	-0.626	0.5334	
$\text{Dlnaturalgasq}_{t-3}$	-2.76	0.0074	
$\text{Dlnaturalgasq}_{t-4}$	0.247	0.8056	
Dlgoldq_{t-1}	0.684	0.4966	
Dlgoldq_{t-2}	-1.00	0.3204	
Dlgoldq_{t-3}	1.26	0.2111	
Dlgoldq_{t-4}	-0.876	0.3844	*Not reject H_0
Granger Causality Test with $\Delta lcrudeoilq$	**t-value**	**t-prob**	**Decision**
Dlcrudeoilq_{t-1}	0.886	0.3787	
Dlgoldq_{t-1}	3.16	0.0023	*Reject H_0
Granger Causality Test with $\Delta lcopperq$	**t-value**	t-prob	**Decision**
Dlcopperq_{t-1}	1.41	0.1633	
Dlgoldq_{t-1}	3.11	0.0026	*Reject H_0

Table 5.2

Thus, from the table, H_0 in tests with $\Delta lcrudeoilq$ and $\Delta lcopperq$ are rejected, and H_0 in test with $\Delta lnaturalgasq$ is not rejected, which implies lcrudeoilq and lcopperq can Granger-cause lgoldq and lnaturalgasq fails to Granger-cause lgoldq.

b.) Causality is meaningful in many areas such as macroeconomics and policy analysis, as it allow to predict the joint distribution of one variable using information set available in another variable. Formally, if the joint distribution of a subset of observable variables is altered by eliminating the history of other variables, then that second group causes the first (Hendry, 2004). This means the relationship between two variables can be reduced to a linear relationship with residuals as white noise. This is supported by Hendry (2004), as he claimed that if a pair of variables is cointegrated then one of them must Granger-cause the other. Thus, Granger causality test, combined with cointegration test, can be used to test economic theories. For example, Wagner (1980) suggests that the development of an industrial economy will be accompanied by an increased share of public expenditure in GNP, i.e. industrial economy development Granger-causes increased public expenditure. However, Demirbas (1999) tested Wagner's law using Turkey's data during 1950 to 1990, and concluded that cointegration analysis suggests there is no cointegrating relationship between the variables in Turkey, and Granger causality tests also shows no evident supports the causality suggested in Wagner's law, which means that relationship may be a result of spurious regression. Therefore, Granger causality gives an alternative approach to verify plausible economic theories. However, this approach may also be biased at sometimes, as it focuses on the unidirectional causality in statistical perspective.

End of Coursework.